MATTHEW FROSST

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PROFESSIONAL SUMMARY

Computational physicist and PhD candidate with a decade of experience in quantitative modelling, high-performance computing, data-driven research, and machine learning (ML). Proven leader in cross-disciplinary collaboration and strategic project execution with $30M+ in compute allocation managed. Known for turning complex technical insights into actionable strategies. Uniquely positioned to help build the systems that will redefine value in the age of AI. Seeking to leverage advanced computational skills to efficiently drive data-driven decision-making in impactful, fast-paced roles at the intersection of analytics, AI, and strategic decision-making.

EDUCATION

PhD (Computational Astrophysics) — Perth, Australia
University of Western Australia, International Centre for Radio Astronomy Research — *2022 – Nov 2025*

- Led multi-national computational projects involving 150+ million CPU-hours (> $30M USD in value).
- Built statistical and ML tools to model complex physical systems and infer causal dynamics in galaxy formation.
- Designed reproducible workflows, advanced analytics pipelines, and high-impact visual communication for academic and stakeholder audiences.

MSc (Physics, Engineering Physics & Astronomy) — Kingston, Canada
Queen's University — *2019 – 2022*

- First-author research measuring scatter in galaxy diversity with modern cosmological simulations and observations.

BScH (Physics, Engineering Physics & Astronomy) — Kingston, Canada
Queen's University — *2015 – 2019*

EXPERIENCE AND LEADERSHIP

ESO Science Support Discretionary Fund Scholarship Recipient — Munich, Germany
European Southern Observatory — *2025 - present*

- Built probabilistic models using Bayesian inference and simulation-to-observation pipelines to match real-world telescope data to theoretical predictions from state-of-the art simulations of the local Universe.
- Advised cross-national teams on data integrity, model fidelity, and risk quantification under uncertainty.

PhD Candidate Representative — Perth, Australia
International Centre for Radio Astronomy Research, University of Western Australia — *2024 - 2025*

- Advocated for students in senior staff strategic meetings; secured $1,200 stipend raise and $45,000 publication fund.
- Drove governance reform to align performance metrics with research quality and long-term impact.

Chair, Computational-Theory Group — Perth, Australia
International Centre for Radio Astronomy Research, University of Western Australia — *2022 - 2024*

- Directed a 15+ member research unit applying supercomputing and machine learning to cosmological questions.
- Designed and led technical and non-technical events, from workshops to public-facing science communication.

Consultant Data Scientist — *Kingston, Canada*
Queen's University — *2017 - 2019*

- Delivered analytics dashboards and reporting pipelines for student performance and resource optimisation.
- Presented data insights to stakeholders including department heads, leading to measurable improvements in retention and instructional quality.

TECHNICAL STRENGTHS, ACTIVITIES, AND INTERESTS

Programming	Python (numpy, pandas, scikit-learn, matplotlib), R, C++, Fortran, SQL
Infrastructure	GNU/Linux, Git, HPC systems, Slurm, Bash scripting, containerization
Methods	Machine learning, Bayesian inference, MCMC, dimensionality reduction, simulation design
Communication	LaTeX, technical writing, stakeholder briefings, public science outreach
Selected Activities	Local Organizing Committees (LOC) for 3 national science events; community outreach
Personal Interests	Ironman triathlon, chess, astronomy education, AI ethics

– References available upon request. –